Project Report: Simplex Architecture for Inverted Pendulum

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1 Linearization

The inverted pendulum has the following equilibrium equaitons:

$$\begin{cases} (m+M)\ddot{x} + \frac{1}{2}ml\,\ddot{\theta} = F - B_x\dot{x} \\ \frac{1}{2}ml\,\ddot{x} + \frac{1}{3}ml^2\ddot{\theta} + \frac{1}{2}mgl\,\theta = -B_\theta\dot{\theta} \end{cases}$$

Given the initial states, we can write the states near the equilibrium like

$$\tilde{x} = x - x_s, \qquad \tilde{\theta} = \theta - \pi, \qquad \dot{\tilde{x}} = \dot{x}, \qquad \dot{\tilde{\theta}} = \dot{\theta}, \qquad \tilde{F} = F - F_s$$

Also, considering the small angle approximations

$$\sin(\theta) = \sin(\pi + \tilde{\theta}) \approx -\tilde{\theta}, \qquad \cos(\theta) = \cos(\pi + \tilde{\theta}) \approx -1$$

We can linearize the system using taylor expansion and get

$$\begin{bmatrix} m+M & \frac{1}{2}ml \\ \frac{1}{2}ml & \frac{1}{3}ml^2 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} B_x & 0 \\ 0 & B_\theta \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2}mgl \end{bmatrix} \begin{bmatrix} x \\ \theta \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

if we isolate the $\ddot{x}$ and $\ddot{\theta}$:

$$\ddot{x} = \frac{4}{4M+m}F - \frac{4B_x}{4M+m}\dot{x} + \frac{6B_\theta}{l(4M+m)}\dot{\theta} + \frac{3mg}{4M+m}\theta$$

$$\ddot{\theta} = -\frac{6}{l(4M+m)}F + \frac{6B_x}{l(4M+m)}\dot{x} - \frac{12(M+m)B_\theta}{ml^2(4M+m)}\dot{\theta} - \frac{6(M+m)g}{l(4M+m)}\theta$$

Now, transforming it into the state space equations in the form $\dot{\mathbf{x}} = A\mathbf{x} + Bu$, we get for $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} = \begin{bmatrix} \tilde{x} & \dot{\tilde{x}} & \tilde{\theta} & \dot{\tilde{\theta}} \end{bmatrix}$ and $u = F$

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\dfrac{4B_x}{4M+m} & \dfrac{3mg}{4M+m} & -\dfrac{6B_\theta}{l(4M+m)} \\ 0 & 0 & 0 & 1 \\ 0 & -\dfrac{6B_x}{l(4M+m)} & \dfrac{6(M+m)g}{l(4M+m)} & -\dfrac{12(M+m)B_\theta}{ml^2(4M+m)} \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ \dfrac{4}{4M+m} \\ 0 \\ \dfrac{6}{l(4M+m)} \end{bmatrix}$$

Alsom for the output $y = Cx + Du$ we have

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

1.1 Definition of the States of the Linearized Model

- $x_1 = x - x_s$ represents the deviation of the cart position from its desired set point x_s. When $x_1 = 0$, the cart is exactly at the desired position.
- $x_2 = \dot{x}$ represents the velocity of the cart. This state describes how fast the cart is moving along the horizontal axis.
- $x_3 = \theta - \pi$ represents the angular deviation of the pendulum from the upright equilibrium position. When $x_3 = 0$, the pendulum is perfectly balanced in the upright configuration.
- $x_4 = \dot{\theta}$ represents the angular velocity of the pendulum, describing how fast the pendulum angle is changing.

By defining the states in terms of deviations from the equilibrium point, the linearized model captures the behavior of the system for small perturbations around the operating condition (x_s, π).

2 High Level Controller

The high-level controller operates in a simple mode. It receives the desired state x_d and the current state x of the system as inputs. Its operation consists of computing the difference $x - x_d$ and publishing this value to the controller as the input, so that the output of the HLC is determined based on the deviation between the current and desired states.

$$State = \begin{bmatrix} x(1) - xd \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

3 High Assurance Controller

The high assurance controller consists on just applying the K values projectd by the optimization problem done in $Hac_solver.m$ which will output the K value.

$$K = \begin{bmatrix} -7.44 & -13.54 & 100.99 & 8.88 \end{bmatrix}$$

To obtain this value, we have to solve the follwoing optimization problem

$$\min_{Q,Z} \quad -\log\det(Q) \tag{1}$$

$$\text{s.t.} \quad QA^T + AQ + Z^T B^T + BZ < 0, \quad Q > 0 \tag{2}$$

$$\alpha_k^T Q \alpha_k \leq 1, \quad k = 1, \ldots, q \tag{3}$$

$$\begin{bmatrix} I & b_j^T Z \\ Z^T b_j & Q \end{bmatrix} \succeq 0, \quad j = 1, \ldots, r \tag{4}$$

$$\text{Then} \quad P = Q^{-1}, \qquad Z = -KQ.$$

In this optimization , A and B represent the state variables obtained in the linearization, Q and Z are variables to be found out in the optimization problem and α and b represent the state and input constraints repectively. Together, they form the optimization problem that allow us to find the recovery region P and the values for K

$$\alpha_1 = \begin{bmatrix} \frac{1}{2.2} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \alpha_2 = \begin{bmatrix} -\frac{1}{2.2} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\alpha_3 = \begin{bmatrix} 0 \\ \frac{1}{5} \\ 0 \\ 0 \end{bmatrix}, \quad \alpha_4 = \begin{bmatrix} 0 \\ -\frac{1}{5} \\ 0 \\ 0 \end{bmatrix}$$

$$\alpha_5 = \begin{bmatrix} 0 \\ 0 \\ \frac{6}{\pi} \\ 0 \end{bmatrix}, \quad \alpha_6 = \begin{bmatrix} 0 \\ 0 \\ -\frac{6}{\pi} \\ 0 \end{bmatrix}$$

$$b_{u1} = \begin{bmatrix} \frac{1}{30} \end{bmatrix}, \quad b_{u1_} = \begin{bmatrix} -\frac{1}{30} \end{bmatrix}$$

4 Decision Logic

The decision mechanism determines which controller should be applied by predicting the future system state and verifying whether it remains inside a predefined recovery region.

At the current time t, the system state is $x(t)$ and the control input is $u(t)$. First, the state is projected one control period ahead using the current input $u(t)$, yielding the predicted state $x(t + \tau_c)$.

The predicted state is then evaluated with respect to the recovery region, which is defined by the quadratic constraint

$$x(t + 1\tau_c)^T P x(t + 1\tau_c) \leq 1 \tag{5}$$

where P is a positive definite matrix that characterizes the ellipsoidal recovery region, defined in the optimization problem.

If the condition above is satisfied, the predicted state remains inside the recovery region and the system continues operating with the High-Performance Controller (HPC). Otherwise, if the condition is violated, the predicted state lies outside the recovery region, and the control is switched to the High-Assurance Controller (HAC) to ensure system safety and recover the state to the safe set.

5 Results

This section presents the closed-loop behavior of the inverted pendulum under the proposed simplex architecture. The objective of the evaluation is to verify whether the system can track position references while preserving the pendulum close to the upright equilibrium and switching to the safe controller whenever required.

5.1 States over time

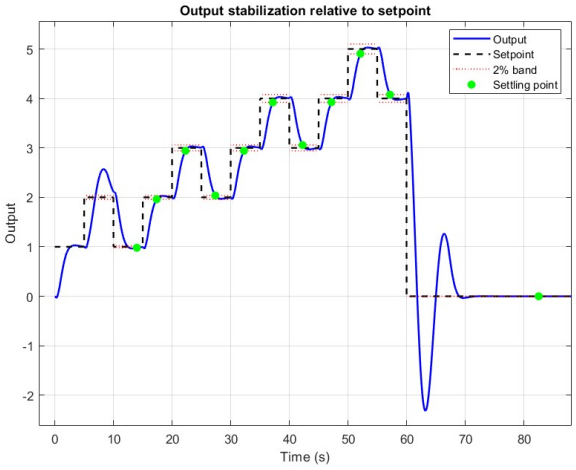


Figure 1: Positions x of the cart over time according to different setpoints over time.

The figures summarizes the evolution of the main state variables during the simulation. The cart position follows the commanded reference changes with a stable transient response, showing that the controllers are able to drive the system toward the requested setpoints without losing stability.

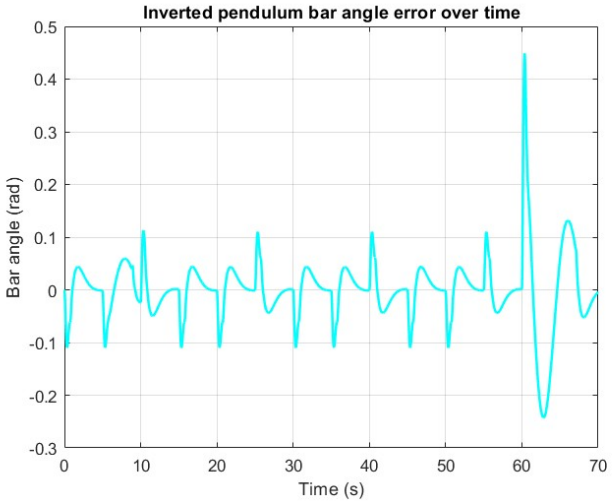


Figure 2: Angle error over time.

In parallel, the angle error remains bounded around zero, which indicates that the pendulum stays close to the upright position even when the cart is commanded to move to a new reference.

The plots also show the expected trade-off between tracking performance and safety. At each reference transition, the cart state presents a transient phase before converging, while the angular deviation increases only temporarily and is then attenuated by the control action. This behavior is consistent with the purpose of the simplex architecture, in which performance is pursued as long as the predicted evolution remains inside the certified safe region.

5.2 HPC acceptance rate

The decision logic activity is illustrated in the following figure, where the switching signal indicates whether the High-Performance Controller or the High-Assurance Controller is active at each instant. This plot provides a direct view of how often the nominal controller is considered safe by the recovery-region test and when the supervisory layer intervenes to maintain safety.

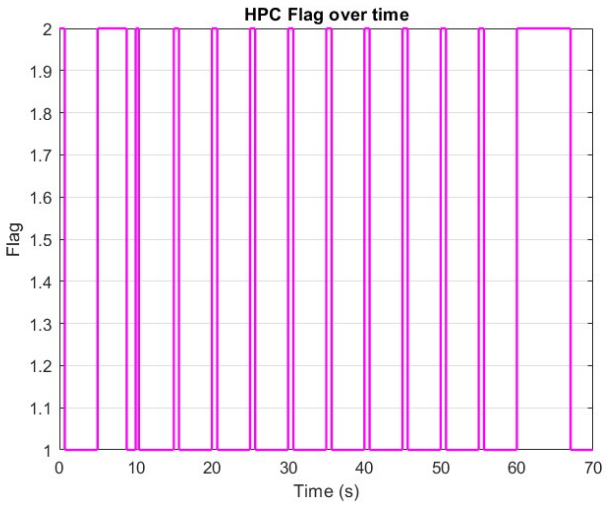


Figure 3: Decision Logic (1 = HPC / 2 = HAC)

The measured HPC acceptance rate was 25.94%. This means that, for roughly one quarter of the simulation time, the predicted state remained inside the admissible region and the high-performance controller was allowed to act. During the remaining intervals, the supervisor selected the HAC, showing that the safety mechanism was frequently required to keep the system inside the Recovery Area.

This percentage was computed from the decision flag signal over a total simulation time of 70 seconds. Considering all sampled points in this interval, the acceptance rate was obtained as

$$extAcceptanceRate = \frac{N_{\text{flag}=2}}{N_{\text{total}}} \times 100\%,$$

where $N_{\text{flag}=2}$ is the number of samples for which the flag is equal to 2, and N_{total} is the total number of sampled points during the simulation. Therefore, the reported value represents the fraction of the full 70-second simulation horizon in which the controller-selection signal assumed the value 2.

Step Time	Reference	Settling Time
10	1	3.98
15	2	2.36
20	3	2.28
25	2	2.36
30	3	2.26
35	4	2.18
40	3	2.26
45	4	2.18
50	5	2.12
55	4	2.18

Table 1: Settling time for each reference step

Table 1 reports the settling time for each reference change. Most setpoint transitions are settled in approximately 2.1 to 2.4 seconds, which indicates a relatively consistent transient response after the initial maneuver.